Beyond Accuracy: Evaluating and Improving Probability Calibration in Machine Learning Models

GitHub: https://github.com/Swethasri-07/Assignment/tree/main

Introduction

Machine learning models are being used everywhere including healthcare, finance, cybersecurity and many more. Historically, the accuracy, precision, recall, and F1-score of these models are performance metrics. Although these metrics reveal useful information about how well a classification method performs, they do not reveal an equally important property: how reliable the predicted probabilities are.

In many real-world applications, the decisions are not based on the predicted labels but on the confidence values of the prediction. To illustrate, if the model predicts probability of disease 90, the disease should occur 90% of the time. A poorly calibrated model may yield predictions that are either over-confident or under-confident ultimately making harmful decisions.

In this tutorial about probability calibration, we will find out how to make predicted probabilities match real-life probabilities. A study on how calibration techniques affect model performance is done on Random Forest and Logistic Regression Models that are trained on the breast cancer dataset. A machine learning system must go beyond accuracy alone to be deemed trustworthy, the researcher said.

Background and Theory:

→**Classification vs Probability Estimation:** Most machine learning classifiers produce both a predicted class and an associated probability. While classification accuracy indicates whether predictions are correct, probability estimates reflect the model's confidence. However, many models produce poorly calibrated probabilities.

For example, ensemble models such as Random Forest tend to produce extreme probability values, often leading to overconfidence. In contrast, linear models such as Logistic Regression typically provide more stable probability estimates, though they may still exhibit calibration issues.

→**Probability Calibration:** Probability calibration aims to ensure that predicted probabilities correspond to actual likelihoods. A well-calibrated model satisfies the condition that, among all predictions with probability p, approximately p proportion should be correct.

Two widely used calibration techniques are:

- **Platt Scaling (Sigmoid Calibration):** A parametric method that fits a logistic function to map predicted scores to probabilities.
- **Isotonic Regression:** A non-parametric method that fits a monotonic function, offering more flexibility.

→**Evaluation Metrics:** Using the model's calibration performance

- **Calibration Curve:** Intersection of predicted probabilities and observed frequencies.
- **Brier Score:** measures the mean square error between predicted probabilities and observed outcomes

Dataset Description

The Breast Cancer dataset consists of 569 instances and is one of the popular benchmark datasets comprising 30 numerical features. Tumours that can either be malign or benign have to be classified.

The dataset is well-suited for analysis of probability calibration since it presents a balanced classification problem and probabilistic prediction, making it a nice choice for evaluating confidence estimates.

Methodology

The dataset was split into training and testing sets using an 80–20 stratified split. Two models were trained:

- Random Forest Classifier
- Logistic Regression

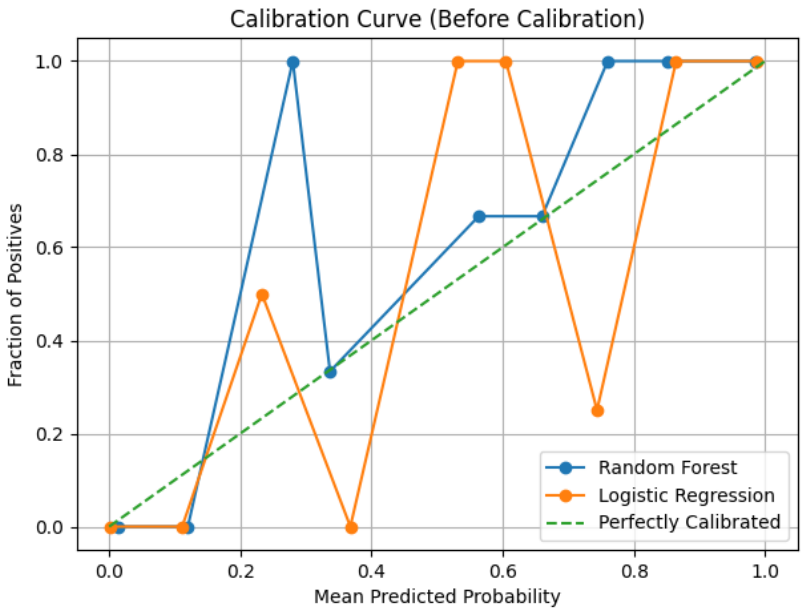
Predicted probabilities were obtained from both models. Calibration techniques were then applied:

- Isotonic regression for Random Forest
- Sigmoid scaling for Logistic Regression

The models were evaluated before and after calibration using calibration curves, Brier scores, and additional visual analysis.

Results

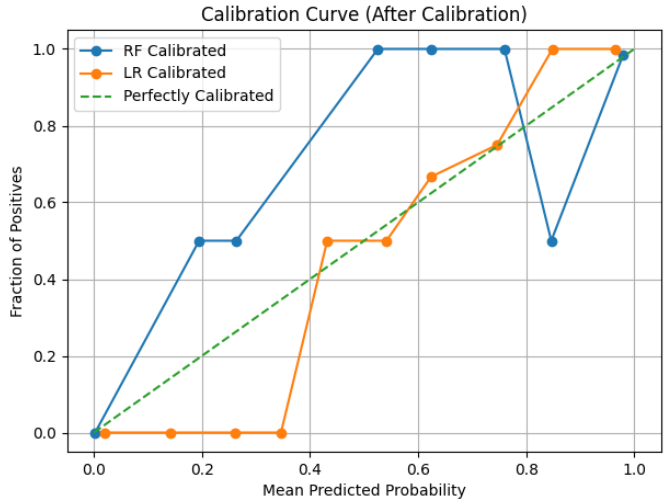
Calibration Before Adjustment:



Before calibration, the Random Forest model deviates significantly from the ideal diagonal line, indicating that its predictions are overconfident. Logistic Regression performs better but still shows minor misalignment. This demonstrates that high classification accuracy does not necessarily imply reliable probability estimates.

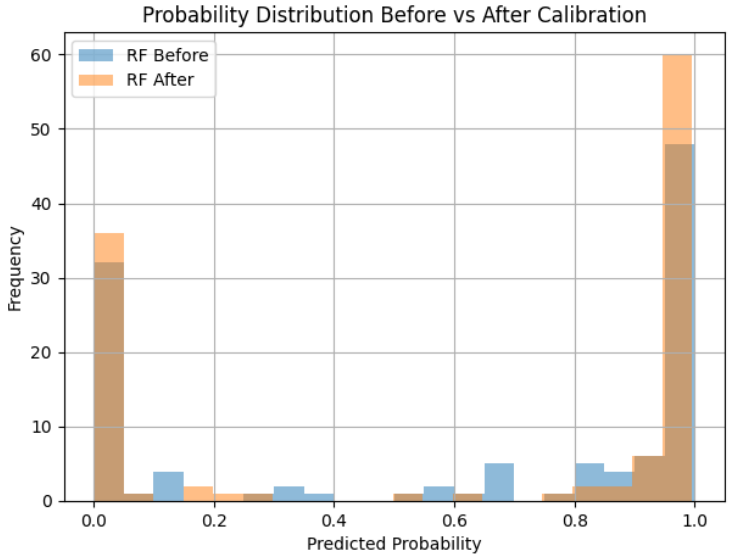
Calibration After Adjustment:

After calibration, both models exhibit improved alignment with the ideal diagonal. The Random Forest model shows a significant improvement, with predictions now better reflecting actual outcome frequencies. This confirms that calibration techniques effectively improve probability reliability.

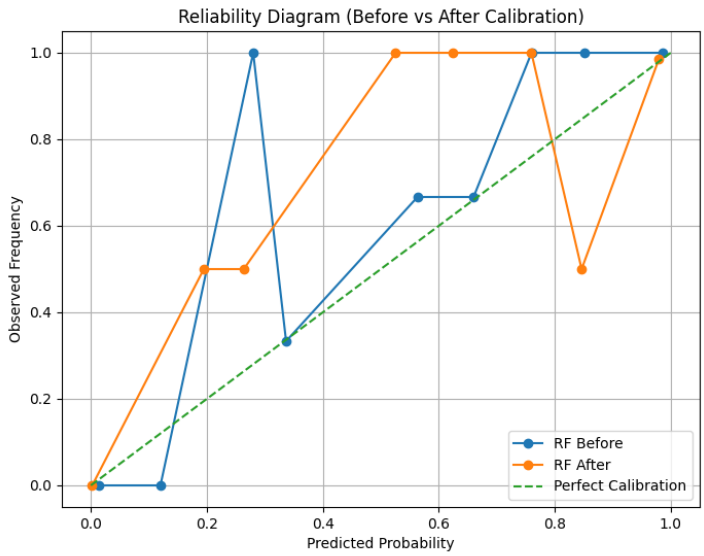


Probability Distribution Analysis:

The probability distribution reveals that, before calibration, predictions are concentrated near extreme values, indicating overconfidence. After calibration, the distribution becomes more balanced, suggesting that the model produces more realistic confidence estimates.

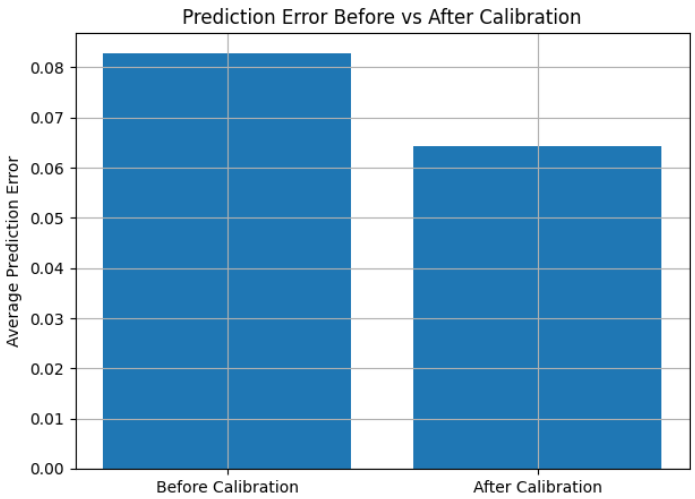


Reliability Analysis:



The reliability diagram provides further insight into calibration performance. Before calibration, predictions deviate from the ideal line, particularly at higher probability levels. After calibration, the curve closely follows the diagonal, indicating improved reliability and reduced bias.

Prediction Error Comparison:



The error comparison plot shows a clear reduction in prediction error after calibration. This indicates that calibration not only improves probability alignment but also enhances overall prediction consistency.

Discussion:

The study found that traditional model evaluation is limited because just measuring accuracy does not provide a good picture of model reliability. The estimates of probabilities for the two were noticeably different in quality, while both performed well.

Random Forest was highly overconfident especially in high probability predictions. This is a common trait of ensemble models, which frequently have extreme outputs. Despite being more stable, Logistic Regression exhibited slight calibration issues.

The probability estimates improved greatly with calibration techniques. Isotonic regression, in particular, was useful to correct the predictions of Random Forest. The reliability diagrams and calibration curves clearly show the alignment with the ideal diagonal improving after calibration.

The calibration's effect on prediction is also clear from the outcome of the probability distribution analysis. The results show calibration reduces prediction extremes and yields heavy-tailed outputs. Moreover, the decrease in prediction error indicates that calibration improves model consistency.

These improvements are significant from a practical view. In practice such as fraud detection or medical diagnosis, decisions are taken based on probability thresholds and not on binary classification. Miscalibration may lead to wrong risk scenarios and poor decisions.

Nonetheless, calibration has its limitations. With this comes added computing complexity and sufficient data for credible estimation. If calibration is improperly applied, it can lead to overfitting.

Probably calibration can improve the reliability of machine learning models and should be considered a core component for evaluating machine learning models.

Ethical Considerations

Calibration is important for ethical Artificial Intelligence. When models are ill-calibrated, confidence in wrong predictions can lead to unnecessary harm. For instance, in the case of medical treatment, such a confident forecast may cause incorrect treatment.

Calibration enhances transparency, fairness, and trust in machine learning systems by improving the reliability of probabilities. In high-stakes applications with high impact decisions, this is especially important.

Conclusion:

This tutorial explored probability calibration and its importance in machine learning. The key findings are:

- Accuracy alone is not sufficient for evaluating model performance
- Random Forest models tend to produce overconfident predictions
- Calibration techniques significantly improve probability reliability
- Visual tools such as calibration curves and reliability diagrams are effective for analysis

Calibration improves both the interpretability and trustworthiness of machine learning models, making it a critical component of real-world deployment.

References

Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow.

Niculescu-Mizil, A., & Caruana, R. (2005). Predicting Good Probabilities with Supervised Learning.

Scikit-learn Documentation – Calibration Module

UCI Machine Learning Repository – Breast Cancer Dataset